Radical Operations and Equations

Concept

Simplify radicals by extracting perfect powers. Check equations after squaring.

Like radicals

Combine only matching radicals after simplification.

Domain

A real square root needs a non-negative radicand.

Rationalize

Multiply numerator and denominator by a radical or conjugate.

Equation steps

Restrict → isolate radical → square → solve → check.

Example 1

Simplify $\sqrt{50} + 2\sqrt{8} - \sqrt{18}$.

Solution

1. Extract squares: $5\sqrt{2} + 4\sqrt{2} - 3\sqrt{2}$.

2. Combine: $6\sqrt{2}$.

Radical Operations and Equations

Example 2

Solve $\sqrt{(x + 6)} = x$.

Solution

1. The right side must be non-negative: $x \geq 0$.
2. Square: $x + 6 = x^2 \rightarrow (x - 3)(x + 2) = 0$.
3. Candidates: 3 and -2. Check original: only $x = 3$ works.

Example 3

Rationalize $1/(\sqrt{3} + 1)$.

Solution

1. Multiply by $(\sqrt{3} - 1)/(\sqrt{3} - 1)$.
2. Result $= (\sqrt{3} - 1)/(3 - 1) = (\sqrt{3} - 1)/2$.

Radical Operations and Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify $\sqrt{108}$.

2. Simplify $3\sqrt{12} - \sqrt{27}$.

3. Rationalize $5/\sqrt{2}$.

4. Solve $\sqrt{(2x + 3)} = 5$.

5. Solve $\sqrt{(x + 2)} = x$.

Radical Operations and Equations

Practice answers | check after trying

1. $6\sqrt{3}$.

2. $6\sqrt{3} - 3\sqrt{3} = 3\sqrt{3}$.

3. $5\sqrt{2}/2$.

4. $2x + 3 = 25$ gives $x = 11$; it satisfies the original.

5. $x \geq 0$. Squaring gives $x^2 - x - 2 = 0$, with candidates 2 and -1. Only 2 satisfies $\sqrt{4} = 2$; -1 is extraneous.